



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN

A Project Proposal On
CAMPUS COMPLAINT MANAGEMENT SYSTEM

Submitted To:
DEPARTMENT OF IT
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN, SUNSARI,
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Table 4.1: Cost-Benefit Analysis of the Proposed Project

1. Introduction

Educational institutions receive a continuous stream of complaints from students regarding campus facilities such as classrooms, laboratories, hostels, sanitation, drinking water, electricity, and internet connectivity. Traditionally, these complaints are lodged in person or through informal channels such as verbal requests to staff, which often leads to complaints being lost, delayed, or left unresolved. Complaint management is essential to ensure that reported issues are properly tracked, monitored, and resolved in a timely manner, as manual handling makes it difficult to follow up on complaints effectively.

This project proposes a mobile-based Complaint Management System that enables students to digitally submit complaints about campus facilities, monitor the status of their complaints in real time, and receive notifications when issues are resolved. The application will be developed using React Native for the frontend, providing a cross-platform mobile experience for Android and iOS users. Firebase will be used for backend services such as user authentication, notifications, and real-time communication, while MongoDB will serve as the primary database for securely storing complaint records, user information, and system data. This combination of technologies provides a modern, scalable, and user-friendly platform that enhances communication between students and the administration, ensuring complaints are managed efficiently and transparently.

2. Problem Statement

At present, students do not have a centralized digital channel to report problems related to campus facilities. Complaints are typically made verbally or through informal notices, which are difficult to track, easy to lose, and often go unanswered without any record of what was reported or when. This lack of a formal system results in:

- No record of what complaints have been raised, by whom, and when.
- No way for students to check the status of a previously submitted complaint.
- Delayed or inconsistent response from the concerned departments, since complaints are not routed systematically.
- No data for the campus administration to identify recurring facility issues and prioritize maintenance.

The proposed Complaint Management System aims to resolve these gaps by giving students a single online platform to report facility issues and giving administrators the tools to track, assign, and resolve them efficiently.

3. Objectives

The primary objectives of this project are:

1. To design and develop a web-based platform where students can submit complaints regarding campus facilities.
2. To allow students to track the real-time status of their submitted complaints (e.g., Pending, In Progress, Resolved).
3. To provide the campus administration with a dashboard to view, categorize, assign, and resolve complaints.
4. To notify students automatically once their complaint status changes.
5. To maintain a historical record of complaints that can help identify recurring facility problems.

4. Methodology

This section outlines the approach and methods that will be used to achieve the project objectives.

4.1. Requirement identification

Requirements for the system were identified by reviewing similar complaint management systems developed for educational institutions, along with a discussion of the specific facility-related issues faced by students on campus.

4.1.1. Study of Existing System / Literature Review

Several studies have explored digital complaint management for institutions. Uwah and Etim [1] describe a web-based student complaint and review system aimed at closing the communication gap between students and academic administration, allowing students to submit complaints or feedback about campus facilities, faculty behaviour, or academic issues through an intuitive interface. Balakrishnan et al. [2] proposed a similar online complaint management platform that uses image recognition to automatically detect the reported problem from an uploaded photo and location APIs to identify where the issue occurred, aiming to close the communication gap between citizens and the responsible authority.

These systems demonstrate the general strengths of digital complaint handling: centralized records, status tracking, and faster routing to the responsible party. Their common limitation, however, is that they are either designed for municipal/citizen use rather than an academic campus, or they do not provide a role-based dashboard tailored to a university's departmental

structure. The proposed system will build on these ideas by tailoring the categories, routing, and dashboard specifically to the university's departments (e.g., hostel administration, IT department, maintenance section).

4.1.2. Requirement Analysis

Based on the above study, the following functional and non-functional requirements were identified:

- Student registration and login using their campus/roll details.
- A complaint submission form with facility category, description, location, and optional photo attachment.
- Status tracking (Pending / In Progress / Resolved) visible to the student.
- An admin dashboard to view, filter, assign, and update complaints.
- Notifications to students when their complaint status changes.
- Non-functional: the system should be responsive (usable on mobile and desktop), secure (authenticated access), and reasonably fast to load.

4.2. Feasibility Study

4.2.1. Technical

The system will be developed using **React Native** for the mobile application frontend, **Firebase** for backend services such as user authentication, cloud messaging, and real-time data synchronization, with **MongoDB** serving as the primary database for storing complaint records and user information. These technologies are open-source or provide free-tier services, are well-documented, and have strong community support, making them suitable for developing a reliable and scalable application. The development team already has familiarity with JavaScript-based technologies, which reduces the learning curve and development time. Additionally, Firebase's free-tier services and MongoDB Atlas's cloud database hosting make the project technically feasible within the available resources and budget.

4.2.2. Operational

Operationally, the system is designed to be simple enough for both students and non-technical administrative staff to use, since the workflow closely mirrors the existing informal complaint process (submit an issue, wait for it to be addressed) but adds transparency and tracking. Adoption will depend on the campus administration being willing to use the admin dashboard

to manage incoming complaints, which is a reasonable expectation given the reduced manual workload it offers compared to handling complaints informally.

4.2.3. Economic

As this is an academic project, development is carried out by the student team without an external contract cost. The main ongoing costs are hosting and domain registration, which can be kept minimal using free or low-cost tiers during development. The table below outlines the cost-benefit structure at a high level; exact figures should be filled in once specific hosting/deployment choices are finalized.

Item	Description	Cost (NPR)	Benefit (NPR)
Development Cost	Developer time (React Native, Firebase, MongoDB)	To be estimated	-
Firestore Services	Firestore Authentication, Cloud Messaging, and other backend services (free tier initially)	To be estimated	-
Database & Hosting Cost	MongoDB Atlas cloud database and application deployment	To be estimated	-
Maintenance Cost	Bug fixes, updates, backups	To be estimated	-
Reduced Manual Effort	Time saved from paper-based complaint handling	-	To be estimated
Improved Student Satisfaction	Faster response and transparent tracking	-	To be estimated
Better Facility Management	Data-driven maintenance decisions for the campus	-	To be estimated

4.2.4. Schedule (Gantt chart showing the project timeline)

The project is planned to be completed within 12 weeks, as shown in the Gantt chart below.

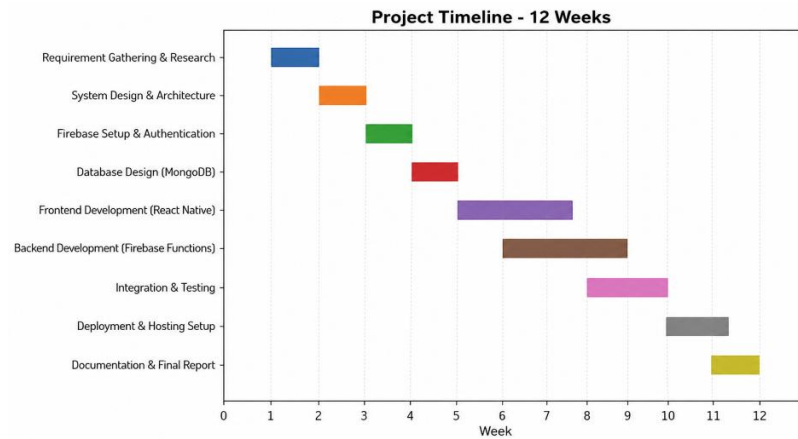


Figure 4.2: Gantt Chart demonstrating schedule feasibility (12 weeks)

4.3. High-Level Design of System

4.3.1. Methodology of the proposed system

The system will be developed using an Agile, iterative development methodology and will follow Object-Oriented Programming (OOP) principles to ensure modularity, maintainability, and scalability. The frontend will be built using React Native, providing a cross-platform mobile application for Android and iOS. Firebase will handle backend services such as user authentication, push notifications, and real-time data synchronization, while MongoDB will be used as the primary database to store information related to users, complaints, departments, and complaint status. This architecture enables efficient communication between the mobile application, Firebase services, and the MongoDB database, resulting in a secure, responsive, and scalable complaint management system.

4.3.2. Flow Charts/Working Mechanism of Proposed System

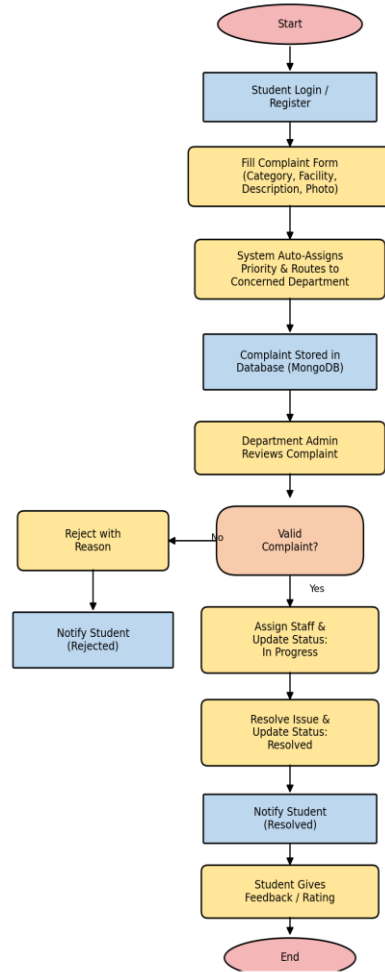


Figure 4.1: Flowchart of the Complaint Management System workflow

4.3.3. Description of Algorithms

A rule-based complaint routing and prioritization algorithm will be used to automatically direct each complaint to the correct department and assign it an initial priority level:

- Step 1: When a complaint is submitted, the system reads its selected category (e.g., Hostel, Electricity, Internet, Sanitation, Classroom).
- Step 2: The category is mapped to the responsible department using a predefined lookup table (e.g., Electricity -> Maintenance Section).
- Step 3: A priority score is assigned based on keyword matching in the description (e.g., words like "leak", "no power", "safety" raise the priority) and the number of similar unresolved complaints already logged for that facility.
- Step 4: The complaint is inserted into the assigned department's queue, ordered by priority score and submission time, so higher-priority and older complaints are shown first to the admin.
- Step 5: When the admin updates the status, the student is notified and the record is timestamped for future reporting.

This approach is intentionally simple (rule-based rather than machine-learning based) so that it is transparent, easy to maintain, and realistic to implement within the project timeline, while still meaningfully improving on the current unordered, manual process.

5. Expected Output

Upon completion, the project is expected to deliver a fully functional web application where students can register, submit complaints about campus facilities with supporting details, and track the status of each complaint through to resolution. The campus administration will have a dashboard to view, assign, and resolve incoming complaints, along with a historical record that can be used to identify recurring facility problems. Overall, the system is expected to reduce the time taken to report and resolve facility issues, improve transparency between students and the administration, and provide data that can support better facility maintenance decisions on campus.

REFERENCES

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